

Adopted Levels: not observed

$S(n)=24210$ syst; $S(p)=-60$ syst; $Q(\alpha)=-9610$ syst [2021Wa16](#)

$\Delta S(n)=500$, $\Delta S(p)=360$, $\Delta Q(\alpha)=350$ (syst, [2021Wa16](#)).

$S(2p)=-2510$ 300, $Q(\epsilon)=16110$ 360, $Q(\epsilon p)=16990$ 300 (syst, [2021Wa16](#)).

Isotope discovery: none.

$S(2p)$ is negative and $S(p)$ may be consistent with zero. ^{34}Ca is a candidate for ground-state two-proton emitter.

Theoretical calculations:

[2026Li03](#): $S(p)=401$ and $S(2p)=-1987$ from an average of Finite-Range Droplet Model (FRDM12), Koura-Tachibana-Uno-Yamada (KTUY) model, Weizsacker-Skyrme (WS4) model, and Duflo-Zuker (DZ31) model combined with the radial basis function mirror symmetry (RBFms) approach.

[2024Li59](#): $S(2p)=-1044$, $T_{1/2}=177.8$ ns. The $Q(2p)$ decay energy was determined using an empirical linear relationship. The $T_{1/2}$ was calculated by inputting the refined decay energy into the new Geiger-Nuttall formula.

[2024Ji17](#): $T_{1/2}=8.1$ fs from an extension of the Bayrak model with and without the deformation term. Used $Q(2p)=2510$ from [2021Wa16](#).

[2023Me08](#): $T_{1/2}=64.6$ ps. The WKB approximation is applied to a Yukawa-plus-exponential model potential. Used $Q(2p)=1470$ from [2017Wa10](#) and a spectroscopic factor derived from a specific cubic-polynomial relation.

[2022Zo01](#): $S(p)=259$ 96, $S(2p)=-2132$ 151, mass excess=14497 98. Mass relations of mirror nuclei.

[2022Sa41](#): $T_{1/2}=2.2$ fs from the Coulomb and proximity potential model for deformed nuclei (CPPMDN) incorporating ground-state deformation (β_2) and orientation effect. Used $Q(2p)=2510$ from [2021Wa16](#).

[2022Ro15](#): $T_{1/2}=0.35$ fs from the generalized liquid drop model (GLDM). $T_{1/2}=3.0$ as from a semi-empirical analytic expression based on a modified version of the Geiger-Nuttall law. Both calculations used $Q(2p)=2510$ from [2021Wa16](#).

[2022Pa24](#): $T_{1/2}=0.31$ ns from a two-potential approach; the nuclear potential is obtained using the Skyrme-Hartree-Fock approach and the Skyrme effective interaction of SLy8. Used $Q(2p)=1470$ from [2017Wa10](#).

[2021Xi06](#): $T_{1/2}=3.5$ fs from the Unified Fission Model (UFM), $T_{1/2}=1.7$ fs from the Generalized Liquid Drop Model (GLDM), $T_{1/2}=27.5$ fs from the Effective Liquid Drop Model (ELDM). These models share a similar physical mechanism by treating the decay as a ^2He cluster preformed on the nuclear surface that subsequently penetrates the Coulomb barrier. All used $Q(2p)=2510$ from [2021Wa16](#).

[2021Li27](#): $T_{1/2}=79.4$ ps from the Gamow-like model. Used $Q(2p)=1470$ from [2017Wa10](#).

[2021Li16](#): $T_{1/2}=117$ ps from a two-parameter empirical formula based on the Geiger-Nuttall law. Used $Q(2p)=1470$ from [2017Wa10](#).

[2020Ma35](#): $S(p)=203$ 151, $S(2p)=-2152$ 118, $T_{1/2}=3.6$ ps +51–20. Mass relations of mirror nuclei with local correlations and a four-parameter empirical formula based on the Effective Liquid Drop Model (ELDM).

[2020Cu01](#): $T_{1/2}=19.5$ ps from a generalized liquid drop model (GLDM). Used $Q(2p)=1470$ from [2017Wa10](#).

[2019Sr02](#): $T_{1/2}=2.2$ ns. A four-parameter empirical formula was fitted using the theoretical results of the Effective Liquid Drop Model (ELDM). Used $Q(2p)=1470$ from [2017Wa10](#).

Other decay half-life calculations: [2017Sa70](#), [2004Pf02](#), [2003Gr04](#), [2005Pf01](#), [2003Gr24](#), [2001Gr29](#).

Other structure calculations: [2005Wa15](#), [2005Zh28](#), [2002Zh09](#), [2001Yo12](#), [1998Co30](#), [1997Co19](#), [1997Pa38](#), [1996Sh20](#), [1995La03](#).

Giant-resonance calculations: [2021Li26](#), [2001Sa57](#), [1997Ha57](#), [2003Ma71](#), [2001Sa77](#), [1999Sa22](#), [1998Ha09](#), [1998Ha39](#), [1998Sa28](#), [1997Ha55](#).

Experimental search:

A two-hour search conducted on March 27, 2012, during the [2015Su01](#) experiment at the Radioactive Ion Beam Line in Lanzhou (RIBLL), did not observe any events at ~ 200 ns that could be attributed to ^{34}Ca in the ΔE -ToF spectrum.

 ^{34}Ca Levels

E(level)	J^π	$T_{1/2}$	Comments
0?	0^+	<35 ns	$\%p=?$; $\%2p=?$ $T_{1/2}$: upper limit estimated from expected cross sections of a ^{40}Ca beam fragmentation on a Ni target at LISE, GANIL (1996PoZZ). The authors of 1986La17 provided additional data from their experiment in a private communication 1996PoZZ.